

**Virginia Electric and Power Company
Surry Power Station
5570 Hog Island Road
Surry, Virginia 23883**

February 4, 2020

U. S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D. C. 20555-0001

Serial No.: 20-041
SPS: SCN
Docket No.: 50-280
50-281
License No.: DPR-32
DPR-37

Dear Sir or Madam:

Pursuant to 10CFR50.73, Virginia Electric and Power Company hereby submits the following Licensee Event Report applicable to Surry Power Station Units 1 and 2.

Report No. 50-280, 50-281 / 2019-002-00

This report has been reviewed by the Station Facility Safety Review Committee and will be forwarded to the Management Safety Review Committee.

Very truly yours,



F. Mladen
Site Vice President
Surry Power Station

Enclosure

Commitment contained in this letter: None

cc: U.S. Nuclear Regulatory Commission, Region II
Marquis One Tower, Suite 1200
245 Peachtree Center Ave., NE
Atlanta, GA 30303-1257

NRC Senior Resident Inspector
Surry Power Station

IEZZ
NRR



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOS-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name	2. Docket Number	3. Page
Surry Power Station Unit 1	05000 280	1 OF 3

4. Title
Items Non-Conforming to Design for Tornado Missile Protection

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
12	11	2019	2019	002	00	02	04	2020	Surry Power Station Unit 2	05000 281
									Facility Name	Docket Number
										05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
N	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☒ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
100	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER	
Licensee Contact Stephen C. Newman, Nuclear Engineer III	Telephone Number (Include Area Code) 757-365-3397

13. Complete One Line for each Component Failure Described in this Report										
Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES	
14. Supplemental Report Expected					15. Expected Submission Date			Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date)					☒ No					

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On December 11, 2019, Surry Nuclear Power Plant determined that the certain Unit 1 and 2 Class 1 Structures, Systems, and Components (SSCs) did not conform with the licensing basis for protection against a design basis tornado generated missile.

The affected equipment was declared inoperable and enforcement discretion invoked in accordance with NRC Enforcement Guidance Memorandum (EGM) 15-002, Rev. 1, "Enforcement Discretion for Tornado Missile Protection Noncompliance." Initial compensatory measures were implemented pursuant to the guidance of NRC Interim Staff Guidance DSS-ISG-2016-01, Appendix A, within the time allowed by the applicable Limiting Conditions for Operation and the associated systems were then declared operable, but nonconforming.

Since the condition was part of the original plant design and licensing of the station, there was no specific cause identified.

There were no safety consequences as a result of this event. Actions are underway to address the condition.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Surry Power Station Unit 1	05000-280	YEAR 2019	SEQUENTIAL NUMBER 002	REV NO. 00

NARRATIVE**Background:**

Surry Nuclear Power Station, Units 1 and 2, are Westinghouse 3-Loop Pressurized Water Reactors [EIS: AC] with a licensed maximum power level of 2546 megawatts thermal. Surry Unit 1 received its initial operating license on May 25, 1972, and Surry Unit 2 received its initial operating license on January 29, 1973, based in part on the design as described in Surry Power Station's Updated Final Safety Analysis Report (UFSAR).

As described in Chapter 2.0 of the UFSAR, Class 1 structures and systems, or parts thereof, whose failure might prevent the simultaneous cold shutdown of both reactor units during a loss-of-power incident will withstand by design a tornado missile with the following characteristics and associated effects:

- Missile equivalent to a wooden utility pole 40-foot long, with 12-inch diameter, weighing 50 lbs/ft³, and traveling in a vertical or horizontal direction at 150 mph.
- Missile equivalent to a 1-ton automobile traveling at 150 mph.

A: Plant Operating Conditions Prior to Event:

Units 1/2

Date: December 11, 2019

Power level: 100 percent/100 percent

Mode: N/N

There were no SSCs out-of-service that contributed to this event.

B: Description of Event:

On December 11, 2019, at 1356 [EST], it was concluded that certain Class 1 SSCs are vulnerable to design basis tornado missiles which could render the equipment inoperable and unable to perform its design function. Specifically, this condition affected the following equipment:

1. Component Cooling Water [EIS: CC] piping for the 'A' Spent Fuel Cooling Water system heat exchanger [EIS: DA-HX]. This heat exchanger is vulnerable to a horizontal missile traveling through the fuel building roll-up door.
2. All three (3) Emergency Service Water (ESW) pumps [EIS: BI-P] and their diesel fuel oil supply tank [EIS: DC-TK]. The ESW pumps and diesel fuel oil tank are vulnerable to a horizontal missile penetrating the tornado missile screens [EIS: SCN].
3. Certain Component Cooling Water system pump discharge piping is vulnerable from a vertical missile penetrating the auxiliary building [EIS: NF] roof.
4. The Unit 1 Auxiliary Feedwater (AFW) system pumps [EIS: BA-P] and the pump suction and discharge piping are vulnerable to a missile traveling through the tornado missile screens on the sides and roof of the main steam valve [EIS: SB-V] house. This vulnerability also exists for the Unit 2 AFW system.
5. The Unit 2 Service Water (SW) supply valves [EIS: KE-V] to the river water make-up pump are vulnerable to a vertical missile traveling through the turbine building roof.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Surry Power Station Unit 1	05000-280	2019	002	00

NARRATIVE

The affected equipment was declared inoperable and enforcement discretion invoked in accordance with NRC Enforcement Guidance Memorandum (EGM) 15-002, Rev. 1, "Enforcement Discretion for Tornado Missile Protection Noncompliance." Initial compensatory measures were implemented pursuant to the guidance of NRC Interim Staff Guidance DSS-ISG-2016-01, Appendix A, within the time allowed by the applicable Limiting Conditions for Operation (LCOs) and the associated systems were then declared operable, but nonconforming. An 8-hour ENS notification was submitted [Event Notification 54437].

This event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by Technical Specification and 10 CFR 50.73(a)(2)(ii)(B) as an event or condition that resulted in the nuclear power plant being in an unanalyzed condition that significantly degraded plant safety. These reporting criteria were selected based on the historical nature of the issue.

C: Cause of Event:

This condition was part of the original plant design and licensing of the station. There was no specific cause identified.

D: Safety Consequences:

As documented EGM 15-002 (Rev. 1), tornado missile scenarios that may lead to core damage are very low probability events because safety-related SSCs are typically designed to withstand the effects of tornados. In addition, because plants are designed with redundancy and diversity, the tornado missiles would have to affect multiple trains of safety systems and/or means of achieving safe shutdown.

E: Corrective Action:

For the Surry Power Station, enforcement discretion provided by EGM 15-002 remains in effect until June 10, 2020, at which time modifications, analysis, license amendments, or other actions must be implemented to resolve the missile vulnerabilities.

F: Previous Occurrences:

There have been no previous Licensee Event Reports at Surry Power Station on this issue.

G: Component Failures:

None.